

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

*I am **K.Manjunadha Reddy / 192372081** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: K.Manjunadha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

CO-4 - Assessment Tool-3

course: Internet programming
code: CSAU 305

K. Manjunadha Reddy
142372081

i) Fundamental concepts in web Development

i) XML (Extensible Markup language)

XML is a markup language used to store, organize and exchange data. It allows developers to create their own tags to describe information.

Example:

```
< student >
```

```
  < name > Arun < / name >
```

```
  < age > 20 < / age >
```

```
< / student >
```

In web applications, XML is commonly used for data exchange between different systems, configuration files, and transferring structured information.

ii) JSP (Java Server Pages)

JSP is a server-side technology based on Java used to create dynamic web pages. JSP allows Java code and special tags to be used along with HTML.

iii) MVC Architecture

MVC stands for Model-view-controller. It is a software architecture that separates an application into three parts.

- * Model:- handles data and business logic.
- * View:- Displays information to the user.
- * Controller:- Receives user requests and controls the flow between Model and View.

iv) XSLT:-

XSLT is a language used to transform XML documents into other formats, such as HTML, XML, or text.

For example, an XML file containing student information can be transformed by XSLT into an HTML table for display in a browser.

v) XML schema:-

XML schema, often called XSD (XML Schema Definition), defines the structure and rules of an XML document. It specifies elements, attributes, data types, and their relationships.

vi) JSTL:-

JSTL is a collection of standard tags used in JSP pages. It provides tags for common operations such as loops, conditions, formatting and database-related tasks.

vii) PHP (Hypertext Preprocessor)

PHP is a server-side scripting language mainly used for creating dynamic websites and web applications. PHP code runs on the web server and produces HTML that is sent to the users browser.

viii) Database connectivity

Database connectivity is the process of allowing a web application to communicate with a database. It enables application to insert, retrieve, update, and delete data.

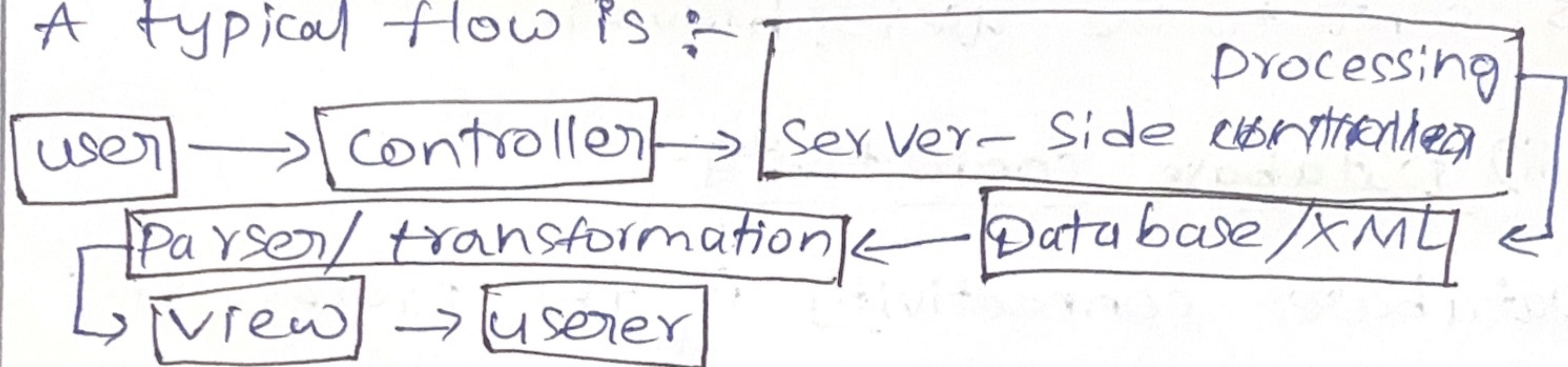
ix) Parser

A parser is a software component that reads and analyzes structured data according to its syntax and structure.

2) Interconnection of these Technologies in a web Application

These technologies can work together as different components of web application.

A typical flow is :-



XML, XML schema, and parser

XML stores data in a structured, hierarchical format. XML schema defines the rules that the XML document must follow.

An XML parser reads the XML document and can check whether it is well-formed and, when used with a schema, whether it satisfies the defined rules.

XSLT and XML

After XML is processed, XSLT can transform the XML data into HTML or another XML format. This is useful when the same XML data needs to be displayed in different ways.

JSP and PHP

JSP and PHP perform server-side processing, when they receive requests from users, execute applications logic, process from data, communicate with database, and generate dynamic responses.

MVC Architecture

MVC provides the overall organization of the application. The controller receives the user's request and decides what action should be performed. The model handles business logic/ and data/database operations. The view, which may be implemented using JSP in a Java application, presents the result to the user.

Database connectivity

The model commonly uses database connectivity to communicate with a database. It can retrieve existing records or store new information. The retrieved data can then be passed to the View for presentation.